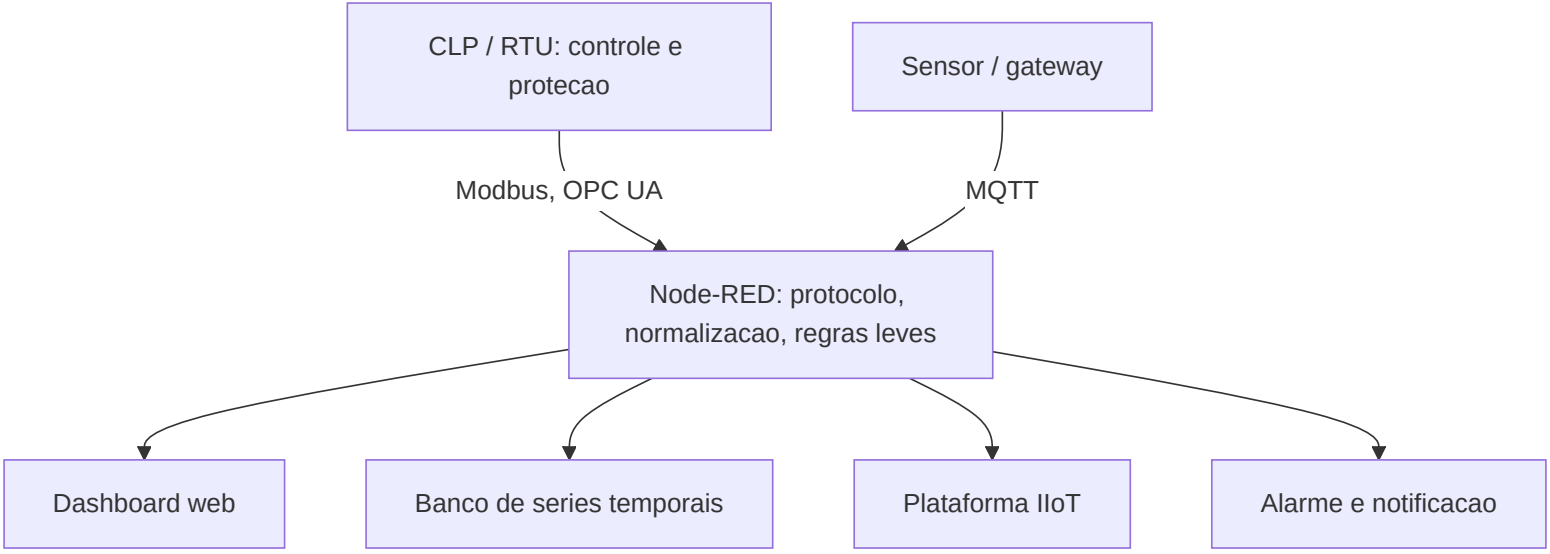
CLP / RTU: controle e protecao
Sensor / gateway
Modbus, OPC UA
MQTT
Node-RED: protocolo, normalizacao, regras leves
Dashboard web
Banco de series temporais
Plataforma IIoT
Alarme e notificacao